Practice workbook · Locate DNA and chromosomes in a human cell

Short items asking where DNA or chromosomes are found, or which molecule or structure carries genetic material.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-LOCATE-GENETIC-MATERIAL

1. Question 4 · part a.i · 1 mark

November 2023 Paper 1

Worked-example retry · Attempt part a.i only.

4 The mass of DNA in a body cell changes during the cell cycle.

(a) (i) Which structure in a cell contains DNA?

(1)

- ☒ **A** chromosome
- ☒ **B** endoplasmic reticulum
- ☒ **C** cell membrane
- ☒ **D** ribosome

(ii) Name **one** other type of nucleic acid.

(1)

.....

(iii) Give **three** differences between DNA and the nucleic acid that you have named in (a)(ii).

(3)

Differences	DNA	Named nucleic acid
1		
2		
3		

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1. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

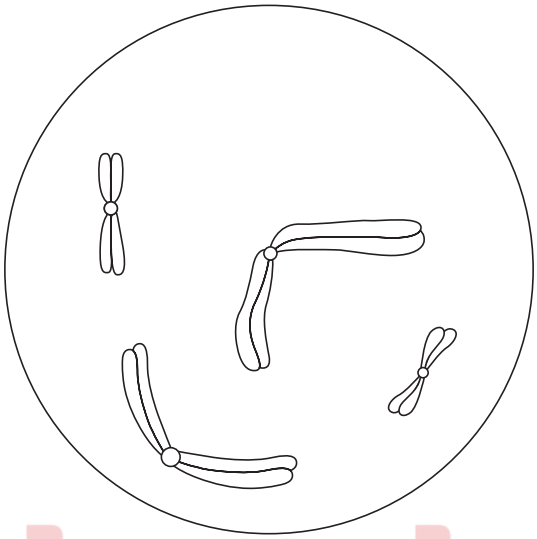
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2. Question 5 · part a.i · 1 mark

June 2021 Paper 2

Further practice · Attempt part a.i only.

5 The diagram shows a cell in the early stages of mitosis.



(a) (i) Which molecule is found in the chromosomes?

(1)

- ☒ **A** DNA
- ☒ **B** glucose
- ☒ **C** glycogen
- ☒ **D** mRNA

(ii) Draw a circle around two homologous chromosomes on the diagram.

(1)

(iii) State how the diagram shows that this is not a human cell.

(1)

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2. Question 5 · part a.i · 1 mark · Working and answer

Working and answer

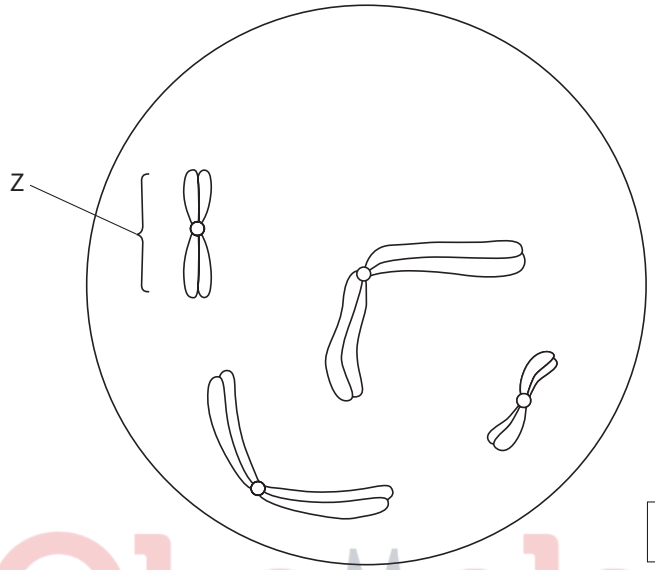
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3. Question 7 · part b.i · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.i only.

7 The diagram shows a cell in early stages of mitosis.



(a) The four stages of mitosis are:

- anaphase
- metaphase
- prophase
- telophase

(i) Give these stages in the correct order.

(2)

.....

.....

(ii) Draw a diagram of the cell as it would appear at metaphase.

(3)

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P 6 7 0 6 0 R R A 0 1 6 2 0

3. Question 7 · part b.i · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.i only.

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(b) (i) Which one of these molecules would be found in structure Z?

(1)

- ☒ **A** DNA
- ☒ **B** mRNA
- ☒ **C** tRNA
- ☒ **D** ribose sugar

(ii) Calculate the length of structure Z in micrometres.

[1 micrometre = 1×10^{-6} m]

(3)

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length of structure Z = ... micrometres

(c) Describe what is produced when a cell undergoes mitosis.

(2)

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(Total for Question 7 = 11 marks)

3. Question 7 · part b.i · 1 mark · Working and answer

Working and answer

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4. Question 1 · part a.iv · 1 mark

June 2022 Paper 1

Further practice · Attempt part a.iv only.

1 (a) The diagram shows part of a molecule of DNA.



(i) Structure X is the backbone of the DNA molecule.

Which of these form part of the backbone of DNA?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

(ii) What name is given to the structures labelled Y?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

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4. Question 1 · part a.iv · 1 mark

June 2022 Paper 1

Further practice · Attempt part a.iv only.

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(iii) Describe how the two backbones of the DNA molecule are joined together.

(2)

(iv) State the name of the part of the cell where most of the DNA is found.

(1)

(b) DNA is a nucleic acid.

RNA is another type of nucleic acid.

(i) Give two differences between the structures of DNA and RNA.

(2)

1

2

(ii) There are different types of RNA.

Which of these are two types of RNA?

(1)

- ☒ **A** mRNA and pRNA
- ☒ **B** dRNA and tRNA
- ☒ **C** dRNA and pRNA
- ☒ **D** mRNA and tRNA

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4. Question 1 · part a.iv · 1 mark · Working and answer

Working and answer

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5. Question 3 · part a.i · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.i only.

3 (a) Polydactyly is an inherited condition.

There are several faulty genes on different chromosomes that can cause polydactyly.

(i) State which part of a cell contains the most chromosomes.

(1)

(ii) Genes exist in alternative forms.

State the name given to an alternative form of a gene.

(1)

(iii) Give one symptom of polydactyly.

(1)

(b) Polydactyly is caused by the dominant form of the gene.

The table shows the genotypes of two parents, R and S.

Parent	Genotype
female (R)	Pp
male (S)	pp

(i) Name the term used to describe the genotype of parent R.

(1)

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5. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

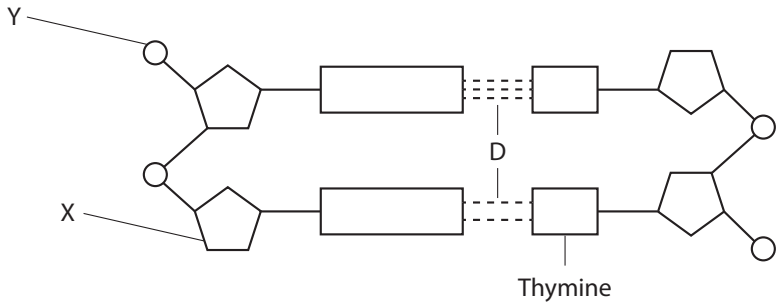
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6. Question 2 · part a.iv · 1 mark

June 2025 Paper 1

Further practice · Attempt part a.iv only.

2 The diagram shows part of a molecule of nucleic acid.



(a) (i) Name the parts labelled X and Y.

(2)

X .....

Y .....

(ii) Name the type of nucleic acid shown in the diagram.

(1)

.....

(iii) Use information from the diagram to justify your answer to part (ii).

(3)

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(iv) State where this nucleic acid is found in a human cell.

(1)

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P 7 8 7 6 8 A 0 4 2 0

6. Question 2 · part a.iv · 1 mark

June 2025 Paper 1

Further practice · Attempt part a.iv only.

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(b) (i) State what is labelled D in the diagram.

(1)

(ii) Describe the importance of D in maintaining the structure of the nucleic acid.

(3)

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(Total for Question 2 = 11 marks)

Working and answer